Agenda

1.) Finish Section 11.10

2.) Start Section 11.11

The Binomial Series

We have a way to expand algebraic expressions of the form $(1+x)^k$ for integer k:

Pascal's triangle

1.) $1+x$

2.) $(1+x)^2$

3.) $(1+x)^3$

4.) $(1+x)^4$

$$(1+x)^k = \sum_{n=0}^{k} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}$$

$$\binom{k}{n} \frac{k!}{n!(k-n)!} = \frac{k(k-1)\dots\overbrace{(k-(n-1))}^{k-n+1}}{n}$$

The Binomial Series allows us to expand this for any exponent k (not just positive integers).

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!} + \dots$$

$$\binom{k}{n} = \frac{k(k-1)\dots(k-n+1)}{n!}$$

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$$

<u>Example:</u> Write $f(x) = \frac{1}{\sqrt{1-x^2}}$ as a power series

$$f(x) = (1-x^2)^{-1/2} = (1+(-x^2))^{-1/2} = \sum_{n=0}^{\infty} \binom{-1/2}{n} (-x^2)^n$$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n \binom{-1/2}{n} x^{2n} = 1 - (-1/2)x^2 + \frac{(-1/2)(-3/2)}{2!} x^4 - \frac{(-1/2)(-3/2)(-5/2)}{3!} x^6$$

$$\frac{3 \cdot 1}{4 \cdot 2} \qquad \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2}$$

$$= 1 + \underbrace{1/2 x^2}_{n=1} + \underbrace{3/8 x^4}_{n=2} + \underbrace{\frac{15}{48} x^6}_{n=3} + \underbrace{\frac{7 \cdot 5 \cdot 3 \cdot 1}{8 \cdot 6 \cdot 4 \cdot 2} x^8}_{n=4} + \underbrace{\frac{9 \cdot 7 \cdot 5 \cdot 3 \cdot 1}{10 \cdot 6 \cdot 4 \cdot 2} x^{10}}_{n=5} + \ldots$$

$$= 1 + \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \ldots (2n)} x^{2n}$$

<u>Taylor Polynomials</u>

<u>Definition:</u> The n^{th} degree Taylor polynomial for $f(x)$ centered at $x=c$ is defined as $T_n(x) = \sum_{i=0}^{n} \frac{f^{(i)}(c)}{i!}(x-c)^i$

$$T_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

Example: Find $T_4(x)$ for $f(x) = \sqrt{1+x}$ centered at $x=\underline{0}$
degree of polynomial — c center

$f(x) = \sqrt{1+x}$ $f(0) = 1$

$f'(x) = \dfrac{1}{2(1+x)^{1/2}}$ $f'(0) = 1/2$

$f''(x) = \dfrac{-1}{4(1+x)^{3/2}}$ $f''(0) = -1/4$

$f'''(x) = \dfrac{3}{8(1+x)^{5/2}}$ $3/8$

$f^{(4)}(x) = \dfrac{-15}{16(1+x)^{7/2}}$ $\dfrac{-15}{16}$

$$T_4(x) = 1 + \tfrac{1}{2}x + \frac{-1/4}{2!}x^2 + \frac{3/8}{3!}x^3 + \frac{-15/16}{4!}x^4$$

$$= 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \tfrac{1}{16}x^3 - \frac{5}{128}x^4$$

The ROC of the Taylor series, $(R=1)$ tells us where this polynomial approximates well.

Activity: Use $\sin x = \sum_{n=0}^{\infty} (-1)^n \dfrac{x^{2n+1}}{(2n+1)!}$ to find $T_6(x)$ for

$g(x) = x\sin x$ centered at $x=0$.

① Find the series representation of $x\sin x$

② Truncate step 1 to find $T_6(x)$

$$x\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{(2n+1)!} = x^2 - \tfrac{1}{3!}x^4 + \tfrac{1}{5!}x^6 - \tfrac{1}{7!}x^8 + \dots$$

$$\boxed{T_6(x) = x^2 - \tfrac{1}{6}x^4 + \tfrac{1}{120}x^6}$$